

```
1  //Lab Code:5555
2  //Jason Johnson
3
4
5  #include <iostream>
6  #include <iomanip>
7  #include <string>
8  #include <fstream>
9
10
11 using namespace std;
12
13 const int SIZE = 31;
14 const int RECORDS = 3;
15
16 struct Students {
17     char name[SIZE];
18     int id;
19     double gpa;
20 };
21
22
23 int main() {
24
25     Students student = { " ", 0, 0.0 };
26
27     fstream data("Data.bin", ios::out | ios::binary);
28
29     cout << "Writing to the file\n";
30
31     for (int i = 0; i < RECORDS; i++) {
32         cout << "\nEnter Information for student " << i + 1 << endl;
33
34
35
36         cout << "Name: ";
37         cin.getline(student.name, SIZE);
38
39
40
41         cout << "ID: ";
42         cin >> student.id;
43
44
45
46         cout << "GPA: ";
47         cin >> student.gpa;
48
49         cin.ignore();
```

```
50
51     data.write((char*)&student, sizeof(student));
52
53
54 }
55 data.close();
56
57 cout << "\nThe size of each student is " << sizeof(student) <<      ↗
    endl << endl;
58
59
60 fstream data1("Data.bin", ios::in | ios::binary);
61
62 cout << "\nHere are the data for all the students\n";
63
64
65 data.read((char*)&student, sizeof(student));
66
67
68 while (!data.eof()) {
69     cout << "\nStudent Name: ";
70     cout << student.name << endl;
71
72     cout << "\nID: ";
73     cout << student.id << endl;
74
75     cout << "\nGPA: ";
76     cout << student.gpa << endl;
77
78     cout << "=====\n";
79
80     data.read((char*)&student, sizeof(student));
81
82
83 }
84 cout << endl << endl;
85
86 data1.close();
87
88 int recordNumber{};
89
90
91
92 fstream file("Data.bin", ios::in | ios::out | ios::binary);
93
94 cout << "Which student record would you like to edit";
95
96 cin >> recordNumber;
97 recordNumber += -1;
```

```
98
99     file.seekg(recordNumber * sizeof(student), ios::beg);
100
101     file.read((char*)&student, sizeof(student));
102
103     cout << endl;
104
105     cout << "Here is the data for student #" << recordNumber + 1 << "
        before editing:\n";
106
107     cout << "\nStudent Name: ";
108     cout << student.name << endl;
109
110     cout << "\nID: ";
111     cout << student.id << endl;
112
113     cout << "\nGPA: ";
114     cout << student.gpa << endl;
115
116     cout << "=====\n";
117
118     cout << "\nEnter the students # " << recordNumber << " new Data\n";
119
120
121     cout << "\nStudent Name: ";
122     cin >> student.name;
123
124     cout << "\nID: ";
125     cin >> student.id;
126
127     cout << "\nGPA: ";
128     cin >> student.gpa;
129
130     cout << "=====\n";
131
132
133     file.seekp(recordNumber * sizeof(student), ios::beg);
134
135     file.write((char*)&student, sizeof(student));
136
137     file.close();
138
139     fstream file1("Data.bin", ios::in | ios::out | ios::binary);
140
141     cout << "\nHere are the new data for all the students.\n";
142
143     file1.read((char*)&student, sizeof(student));
144
145
```

```
146     while (!file1.eof()) {
147         cout << "\nStudent Name: ";
148         cout << student.name << endl;
149
150         cout << "\nID: ";
151         cout << student.id << endl;
152
153         cout << "\nGPA: ";
154         cout << student.gpa << endl;
155
156         cout << "=====\n";
157
158         file1.read((char*)&student, sizeof(student));
159
160
161     }
162     cout << endl << endl;
163
164     file1.close();
165
166
167
168     fstream file2;
169
170     Students s2 = { " ", 0, 0.0 };
171     Students s3 = { " ", 0, 0.0 };
172
173     cout << "Before the swap:\n";
174
175
176     file2.open("Data.bin", ios::in | ios::out | ios::binary);
177
178     file2.seekg(1 * sizeof(student) | ios::beg);
179
180     file2.read((char*)&student, sizeof(student));
181
182     file2.seekg(2 * sizeof(student) | ios::beg);
183
184     file2.read((char*)&student, sizeof(student));
185
186     file2.seekp(2 * sizeof(student), ios::beg);
187     file2.write((char*)&s2, sizeof(student));
188
189     file2.seekp(1 * sizeof(student), ios::beg);
190     file2.write((char*)&s3, sizeof(student));
191
192     file2.close();
193
194
```

```
195     cout << "After Swap Student 2";
196
197     while (!file2.eof()) {
198         cout << "\nStudent Name: ";
199         cout << s2.name << endl;
200
201         cout << "\nID: ";
202         cout << s2.id << endl;
203
204         cout << "\nGPA: ";
205         cout << s2.gpa << endl;
206
207         cout << "=====\n";
208
209         file2.read((char*)&s2, sizeof(student));
210
211     }
212
213
214     cout << "After Swap Student 3";
215
216
217     while (!file2.eof()) {
218         cout << "\nStudent Name: ";
219         cout << s3.name << endl;
220
221         cout << "\nID: ";
222         cout << s3.id << endl;
223
224         cout << "\nGPA: ";
225         cout << s3.gpa << endl;
226
227         cout << "=====\n";
228
229         file2.read((char*)&s2, sizeof(student));
230
231     }
232 }
233
234
235 // NOT FINISHED WILL STUDY THIS MORE :(
236
```